CogniSense: A Hybrid Edge-Cloud Framework for Physiotherapy Exercise Assessment Using Monocular Pose Estimation

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Abstract—Home-based physiotherapy requires consistent exercise execution and timely feedback, yet continuous professional supervision is difficult to scale. Computer-vision-based assessment offers a potential solution, but purely geometric approaches can be sensitive to variations in movement, while cloud-based vision-language model (VLM) inference introduces network latency, operational overhead, and additional privacy considerations. This paper presents CogniSense, a hybrid edge-cloud framework designed to combine client-side pose estimation, kinematic analysis, lightweight classification, selective cloud verification, and tool-grounded coaching. The client-side pipeline uses MediaPipe pose landmarks, adaptive One Euro filtering, a finite-state repetition segmentation machine, and checkpoint-based kinematic scoring. A lightweight logistic regression classifier operates on a 22-dimensional feature representation, while a confidence-based routing mechanism determines whether a repetition is resolved locally or escalated for cloud verification. The backend provides longitudinal exercise analytics and exposes read-only analytical tools through the Model Context Protocol (MCP). The implementation was examined through automated software tests comprising 80 backend tests and 88 frontend tests. An analytical routing example illustrates how the proposed cascade can reduce cloud invocations under specified routing assumptions. However, the current work does not establish independently measured classification accuracy, clinical effectiveness, or generalization to real patient movements. CogniSense is presented as a software and architectural feasibility study that establishes a foundation for future reproducible experiments and clinical validation.

Index Terms—Physiotherapy, telerehabilitation, monocular pose estimation, edge-cloud computing, kinematic analysis, logistic regression, vision-language models, Model Context Protocol.

I. INTRODUCTION

Physiotherapy exercises are frequently prescribed for home execution following injury, surgery, or the management of musculoskeletal conditions. Their effectiveness depends not only on adherence but also on movement quality, range of motion, tempo, and the avoidance of compensatory movement patterns. In unsupervised environments, users may have limited access to immediate corrective feedback between professional consultations.

Computer vision provides a potential means of assessing movement using commodity cameras. Modern pose-estimation frameworks can extract human-body landmarks from video without requiring wearable sensors or dedicated motion-capture equipment. These landmarks can be processed to estimate joint angles, characterize repetition trajectories, and identify deviations from prescribed movement patterns.

However, deploying an exercise-assessment system involves several engineering challenges. First, raw pose landmarks are affected by image noise, occlusion, and variations in camera placement. Second, geometric threshold-based assessment can be difficult to generalize across exercises and movement patterns. Third, sending every repetition to a cloud-hosted vision-language model may introduce unnecessary network requests and dependence on external services. Finally, systems that combine conversational artificial intelligence with exercise analytics require restrictions on the information and operations available to the agent.

CogniSense addresses these challenges through a hybrid architecture that distributes computation across three tiers: an edge client, a cloud application backend, and a cloud intelligence layer. The edge client performs pose processing and initial exercise assessment. The backend manages application data and computes longitudinal analytics. The intelligence layer supports selective visual verification and tool-grounded coaching.

A. Principal Contributions

The principal contributions of this work are:

1) An edge-side kinematic processing pipeline integrating landmark smoothing, joint-angle computation, repetition segmentation, and checkpoint-based movement scoring.

2) A confidence-based edge-cloud decision cascade combining geometric checks, lightweight logistic regression, and selective cloud escalation.

3) A tool-grounded coaching architecture that exposes restricted analytical operations through an MCP server rather than granting unrestricted access to application functions.

4) A software verification and analytical evaluation framework that distinguishes implemented functionality and test coverage from empirical performance and clinical validation.

The current study focuses on system architecture and implementation. It does not claim validation against real patients, clinician annotations, or a clinically established ground-truth dataset.

II. RELATED WORK

A. Monocular Human Pose Estimation

Human pose estimation seeks to identify anatomical landmarks from images or video. Frameworks such as BlazePose and MediaPipe provide computationally efficient approaches for estimating body landmarks on consumer hardware [1], [2]. Their outputs can be used to derive approximate joint angles and movement trajectories without requiring dedicated motion-capture equipment.

Despite these advantages, monocular pose estimation has limitations. Two-dimensional landmark coordinates do not directly provide complete three-dimensional biomechanics. Perspective distortion, out-of-plane movement, occlusion, and camera placement can affect the estimated joint geometry. Consequently, a pose-based assessment system must account for uncertainty and avoid interpreting estimated kinematics as direct clinical measurements.

B. Computer-Assisted Rehabilitation Assessment

Computer-assisted rehabilitation systems have investigated automated assessment of exercise execution using motion data, skeletal representations, and learned movement models. Public datasets, including UI-PRMD and KIMORE, have been used in prior rehabilitation research [3], [4].

These datasets provide relevant background for the development and evaluation of automated exercise-assessment methods. However, the present CogniSense implementation has not been demonstrated to use these datasets in its experimental pipeline. They are therefore discussed as related research resources rather than as data evaluated in this study.

C. Vision-Language Models and Tool-Grounded Agents

Vision-language models can process image inputs and produce natural-language descriptions or judgments. Such capabilities may complement deterministic kinematic processing when a movement requires additional contextual interpretation. However, external model calls introduce network dependencies, potential operational costs, and data-governance considerations.

Tool-using language-model architectures provide a mechanism for separating natural-language interaction from application operations. The Model Context Protocol defines a standardized approach for exposing tools and contextual resources to compatible clients [5]. CogniSense uses a restricted analytical tool layer to limit the operations available to the coaching agent.

III. SYSTEM ARCHITECTURE

CogniSense follows a three-tier architecture comprising the edge client, cloud application backend, and cloud intelligence layer.

A. Edge Client

The client is implemented using React and Vite. It captures camera input through browser media interfaces and processes pose landmarks using MediaPipe. The video-processing pipeline performs the following operations:

1) Extract body landmarks from camera frames.

2) Smooth landmark trajectories using an adaptive One Euro filter.

3) Calculate joint angles and other kinematic quantities.

4) Segment movement into individual repetitions.

5) Evaluate repetitions using geometric checkpoints and derived quality features.

6) Run the lightweight classifier and determine whether further verification is required.

Keeping pose extraction and initial assessment on the client reduces the need to transmit continuous video to the backend. This design does not eliminate all data transmission: the architecture permits selected image snapshots to be sent for cloud verification when configured and authorized.

B. Cloud Application Backend

The backend is implemented using Node.js and Express. Its responsibilities include application request validation, authentication integration, persistence, and longitudinal exercise analytics.

The analytical layer tracks changes in exercise-related measurements over time. Potential indicators include changes in repetition quality, range of motion, tempo, and adherence. Such indicators should be interpreted as software-derived trends rather than clinical diagnoses unless validated through appropriate clinical research.

C. Cloud Intelligence Layer

The cloud intelligence layer integrates a vision-language model for selected verification requests and an MCP server that exposes read-only analytical tools to the coaching agent.

The restricted tool interface separates conversational reasoning from application data access. Authentication, input validation, tool-level authorization, and safety checks are intended to constrain the agent's actions. The current implementation includes seven read-only analytical tools, as documented in the repository audit.

IV. METHODOLOGY

A. Landmark Smoothing

Pose-estimation outputs may contain high-frequency jitter. CogniSense applies the One Euro filter [6] to smooth landmark trajectories while adapting the cutoff frequency to the observed movement speed.

The adaptive cutoff frequency is defined as

$$f_c = f_{min} + beta * |v| \quad (1)$$

where f_min is the minimum cutoff frequency, beta controls the speed-dependent adaptation, and v is the estimated signal velocity.

The smoothing coefficient is

$$alpha = (2 * pi * f_c * dt) / (1 + 2 * pi * f_c * dt) \quad (2)$$

where dt denotes the interval between observations. The filtered coordinate is obtained by combining the current observation with the previous filtered value using this coefficient.

This adaptive formulation aims to suppress small fluctuations during relatively stationary movement while retaining responsiveness during faster transitions. Its effectiveness under different camera conditions and movement patterns requires systematic benchmarking.

B. Joint-Angle Computation

Let three detected landmarks represent points A, B, and C, with B as the joint vertex. Define the vectors:

$$u = A - B, \quad v = C - B \quad (3)$$

The angle at the joint is calculated using the normalized dot product:

$$theta = arccos((u \cdot v) / (||u|| \, ||v||)) * (180 / pi) \quad (4)$$

The implementation must handle missing landmarks, low-confidence coordinates, and near-zero vector magnitudes to avoid invalid numerical results. The resulting angles are used as approximate two-dimensional kinematic measurements. They should not be interpreted as complete three-dimensional joint angles.

C. Repetition Segmentation

CogniSense implements a finite-state machine to segment movement into repetitions. Its principal states are: IDLE (waiting for movement initiation), ASCENDING (tracking initial movement phase), PEAK (identifying target movement region), DESCENDING (tracking return phase), and COMPLETE (finalizing repetition and producing derived measurements). State transitions depend on exercise-specific movement conditions and observed joint angles. Incomplete movements and implausibly short attempts can be rejected by configured rules.

This state-based design provides a deterministic mechanism for organizing continuous landmark streams into discrete assessment units. Its robustness under occlusion, interrupted tracking, and irregular movement requires dedicated testing.

D. Checkpoint-Based Kinematic Scoring

For each completed repetition, the angle trajectory is resampled at five normalized temporal positions: t_k in {0, 0.25, 0.50, 0.75, 1.00}. Let theta(t_k) be the observed angle at checkpoint k, rho_k the corresponding reference angle, and tau_k its permitted tolerance. A checkpoint satisfies the configured geometric criterion when:

$$|theta(t_k) - rho_k| <= tau_k \quad (5)$$

Checkpoint outcomes contribute to the geometric assessment of the repetition. Additional features characterize range of motion, tempo balance, and trajectory smoothness.

A tempo score can be defined using the ratio of ascent and descent durations:

$$T = clip(100 - 45 * |1 - (t_a / t_d)|, 20, 100) \quad (6)$$

where t_a and t_d denote ascent and descent durations.

A smoothness score may be computed from variation in the angle derivative:

$$S = clip(100 - min(60, 1.5 * sigma_dtheta^2), 25, 100) \quad (7)$$

where sigma_dtheta^2 denotes the relevant derivative-variation statistic.

The implementation combines its configured range-of-motion, smoothness, and tempo components into an aggregate quality score. These scores encode programmed movement criteria and should not be treated as clinically validated measures of rehabilitation quality.

E. Lightweight Classification

CogniSense implements logistic regression as a lightweight client-side classifier. Its feature representation contains 22 values: 16 normalized trajectory samples, a normalized peak-angle feature, a duration feature, and four derived kinematic scores.

Let x be the feature vector, mu and s the training-derived feature means and standard deviations, w the learned weights, and b the bias. The estimated probability is:

$$p(y = 1 | x) = sigma(w^T * ((x - mu) / s) + b) \quad (8)$$

where sigma(z) = 1 / (1 + exp(-z)). The classifier is trained using batch gradient descent with regularization, as implemented in the repository. The application can use its output as a confidence signal for routing decisions.

An important methodological limitation is that some classifier features are derived from the same geometric rules used in the assessment pipeline. If synthetic labels are generated from those rules, the classifier may primarily approximate the existing scoring function. High agreement under that setup would not demonstrate independent recognition of physiotherapy correctness. Establishing the classifier's added value requires an independent evaluation against suitable reference labels and a comparison with the underlying rule-based approach.

F. Confidence-Based Edge-Cloud Routing

After a repetition is processed locally, the architecture applies a three-stage decision cascade:

• Stage 1: Geometric gate. Repetitions that violate configured checkpoint conditions can be rejected locally without a cloud request.

• Stage 2: On-device classification. Repetitions that pass the geometric gate are evaluated by logistic regression. High-confidence predictions can be resolved locally, subject to the configured policy.

• Stage 3: Selective cloud verification. Predictions falling within an ambiguous probability interval may be escalated to the cloud VLM. A configured audit sample of otherwise confident predictions can also be selected for verification.

The probability thresholds and audit rate are routing parameters, not guarantees of correctness. Their impact on classification quality, request volume, and latency must be measured experimentally.

For illustration, the routing arithmetic in the design can be expressed using a hypothetical workload of 1,200 repetitions. If 426 are resolved by the geometric gate, 554 additional repetitions are resolved locally, and 220 are escalated, the modeled local-resolution proportion is:

$$R_local = ((426 + 554) / 1200) * 100 = 81.67\% \quad (9)$$

This is an analytical example based on specified counts, not a claim that a 1,200-repetition experimental benchmark was executed. The corresponding cloud-call reduction would equal 81.67% relative to an assumed baseline of one cloud call per repetition, provided each escalated repetition results in exactly one call. Actual bandwidth savings and performance trade-offs require measurement.

G. Tool-Grounded Coaching and Safety

The conversational coaching component accesses analytical functionality through the MCP server. Read-only tools can provide information about exercise history and application-derived measurements without granting the agent unrestricted access to backend operations.

The architecture includes safety filters for selected warning terms, such as sharp pain, numbness, and dizziness. Detection of such terms can interrupt ordinary coaching and direct the user to seek appropriate medical advice.

Keyword-based filters have inherent limitations: they may miss relevant expressions, misunderstand context, or trigger on benign mentions. They are therefore safeguards within the application, not substitutes for clinical oversight or a validated medical safety system.

V. IMPLEMENTATION AND SOFTWARE VERIFICATION

The application combines a React 19 frontend with a Node.js and Express backend. Pose processing, geometric computation, repetition segmentation, and classification are implemented within the client-side pipeline. Backend services support authentication, persistence, analytical operations, and selected cloud-model interactions.

TABLE I. Automated Software Verification Results

Test Suite	Reported Passing Tests	Verification Outcome
Backend Node.js tests	80	Passing (100%)
Frontend Vitest tests	88	Passing (100%)
Total Automated Tests	168	Passing (100%)

Note: Verified via npm test runner offline. Backend tests cover API routes, MCP tools, and schemas; frontend tests cover angle geometry, One Euro smoothing, and offline queues.

These results document the reported outcomes of the software test suites. The backend tests cover application and tool behavior, while the frontend tests cover selected mathematical functions, filtering behavior, and client-side functionality.

Passing automated tests provides evidence that the tested behaviors satisfy their specified assertions. It does not independently establish classifier generalization, clinical accuracy, end-to-end performance under all operating conditions, or the correctness of every architectural assumption.

A reproducible performance evaluation should additionally record software versions, hardware configuration, random seeds, dataset generation parameters, model configuration, test-set construction, and raw evaluation outputs. Such artifacts are necessary before numerical performance claims can be treated as experimental findings.

VI. DISCUSSION

A. Architectural Implications

CogniSense demonstrates an implementation approach in which multiple forms of computation are assigned to different stages of an exercise-assessment pipeline. Pose extraction and basic kinematic analysis are performed locally, while selective cloud verification is reserved for cases designated by the routing policy.

This design provides a mechanism for controlling when external inference is requested. Its practical benefit depends on the distribution of ambiguous cases, the accuracy of local decisions, the cost of external inference, and the latency of the network connection.

The analytical routing example illustrates how local resolution and cloud-call counts can be calculated from an explicitly specified workload. It does not establish that the same proportions occur in real usage.

B. Synthetic Labels and Circularity

A central limitation of the current evaluation approach is the relationship between programmed labels and the features used by the classifier. When synthetic labels are determined by angle ranges, checkpoint tolerances, or related scoring rules, and the input features contain those same rule-derived scores, the classification problem may become an approximation of a deterministic function.

This setup can be useful for testing feature extraction, model integration, confidence thresholds, and routing behavior. However, it cannot establish that the classifier recognizes clinically meaningful errors that are absent from the programmed rules.

Future experiments should compare the classifier with the underlying rule-based method and evaluate both on independently generated trajectories. The evaluation should separate training and test trajectory families, document the label-generation procedure, and assess whether the classifier provides measurable value beyond the rules it approximates.

C. Evaluation Status and Reproducibility

The current work does not substantiate a completed independent 1,200-repetition benchmark, an executed four-configuration accuracy ablation, or systematically logged latency distributions. Those results should not be presented as experimentally measured findings.

A future reproducible benchmark should save generated test cases, labels, model predictions, routing decisions, and timing measurements. Classification metrics should be calculated directly from the saved predictions. Ablation experiments should use consistent test cases and clearly defined configurations. Cloud-model evaluations should separately document the model version, request conditions, response handling, and actual measured latency.

D. Privacy and Data Governance

The architecture is designed to keep continuous video processing on the client and to limit cloud transmission to selected verification requests. This is a privacy-oriented design choice, but it does not by itself establish compliance with HIPAA, GDPR, or other regulatory frameworks.

A complete assessment must consider application logging, storage, authentication, transport security, third-party model retention policies, and the actual content of transmitted requests. Any privacy guarantee should be limited to behavior demonstrated by the implementation and the applicable service terms.

E. Limitations

The current work has several limitations:

1) No clinical validation: No real patients were recruited and no clinician-annotated dataset was evaluated.

2) No verified public-dataset experiment: The repository audit did not establish that UI-PRMD or KIMORE data were downloaded and processed by the current evaluation pipeline.

3) Potential label circularity: The classifier uses features partly derived from the rules that can also define synthetic labels.

4) No established independent test benchmark: The reported 1,200-case evaluation target is not supported by a saved, independently partitioned test dataset.

5) Unexecuted performance ablations: The proposed routing configurations have not been substantiated by logged, comparable empirical runs.

6) Incomplete latency profiling: Systematic timing distributions across a documented hardware configuration are not available.

7) Monocular sensing limitations: Two-dimensional pose estimation cannot fully recover three-dimensional movement and may be affected by occlusion and perspective.

8) Unvalidated clinical thresholds: Exercise-specific ranges and compensation rules require appropriate expert review and clinical evaluation before being used to support treatment decisions.

VII. CONCLUSION AND FUTURE WORK

This paper presented CogniSense, a hybrid edge-cloud framework for physiotherapy exercise assessment using monocular pose estimation, kinematic feature extraction, lightweight logistic regression, selective cloud verification, and tool-grounded coaching. The architecture integrates a client-side movement-processing pipeline with a backend for application management and longitudinal analytics, alongside a restricted MCP interface for conversational access to analytical tools.

The repository audit reports 168 passing automated tests across the backend and frontend suites. The documented routing design also permits analytical estimation of local-resolution and cloud-invocation proportions under explicitly stated assumptions. These findings support the implementation and software-verification aspects of the work, but do not establish clinical performance or independent classification accuracy.

Future work will focus on constructing a saved and reproducible synthetic benchmark, separating training and test trajectory families, comparing learned classification against the underlying geometric rules, and executing logged routing and latency experiments. Subsequent research should evaluate the framework using appropriately licensed public datasets and, where ethically approved and feasible, real movement recordings with suitable expert annotation. These steps are necessary to determine whether the architecture improves assessment reliability and efficiency beyond its current software-level demonstration.

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